TERMS OF MOVEMENT

LEARNING OBJECTIVES

At the end of lecture student must be able to:

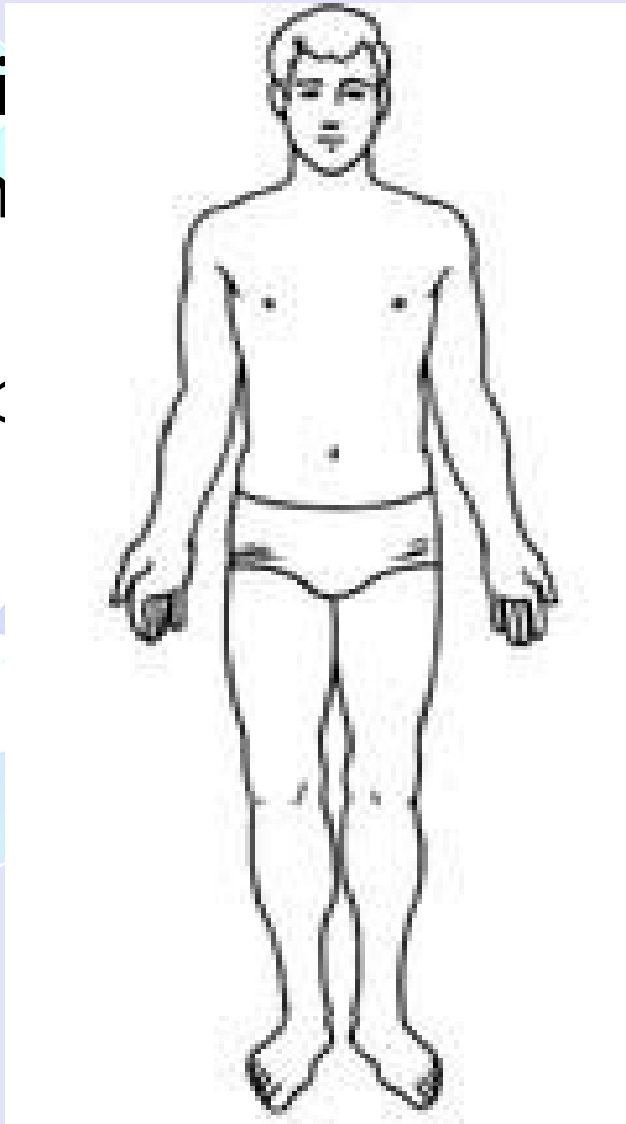
- Define and demonstrate the movements occurring at various joints of body (flexion, extension, abduction, adduction, rotation)
- Identify movements occurring at specific joints (pronation, supination, inversion, eversion)
- Identify the planes at which the movements occur

MOVEMENT

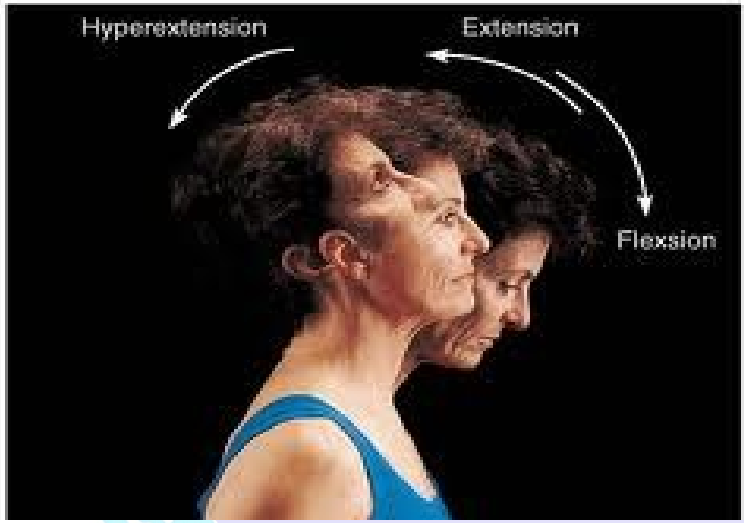
- The movement of body structures is accomplished by the contraction of muscles.
- Muscles may move parts of the skeleton relatively to each other.
- May move parts of internal organs relatively to each other.
- All such movements are classified by the directions in which the affected structures are moved.

ANATOMICAL POSITION

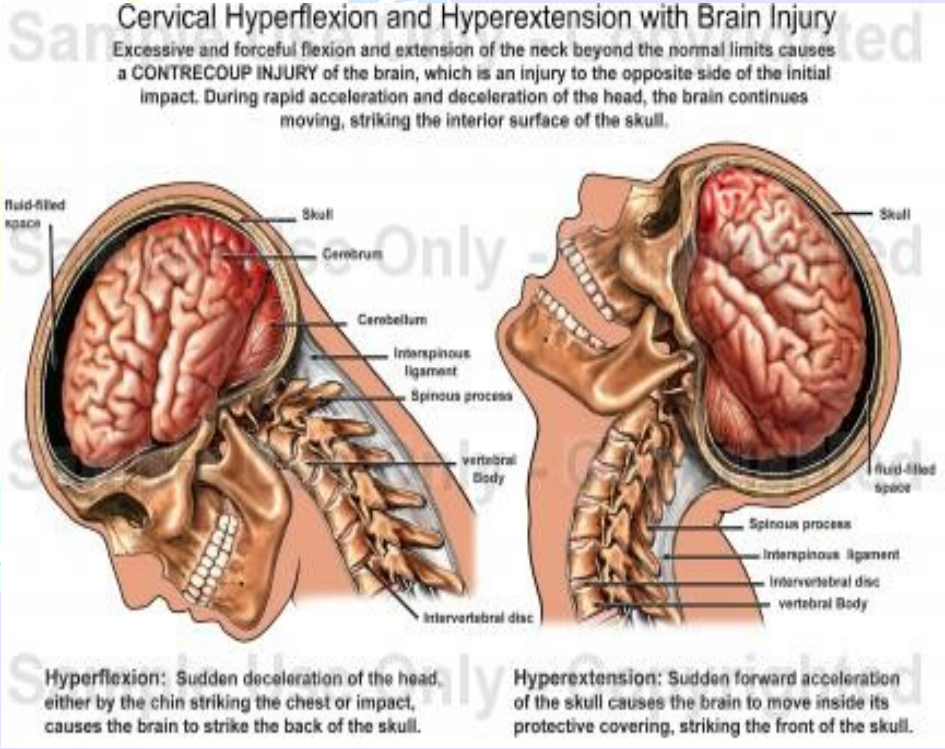
- In human anatomy, all descripti
  and movement are based on th
  that the body is in its complete
  abduction stage in anatomical p



HYPER

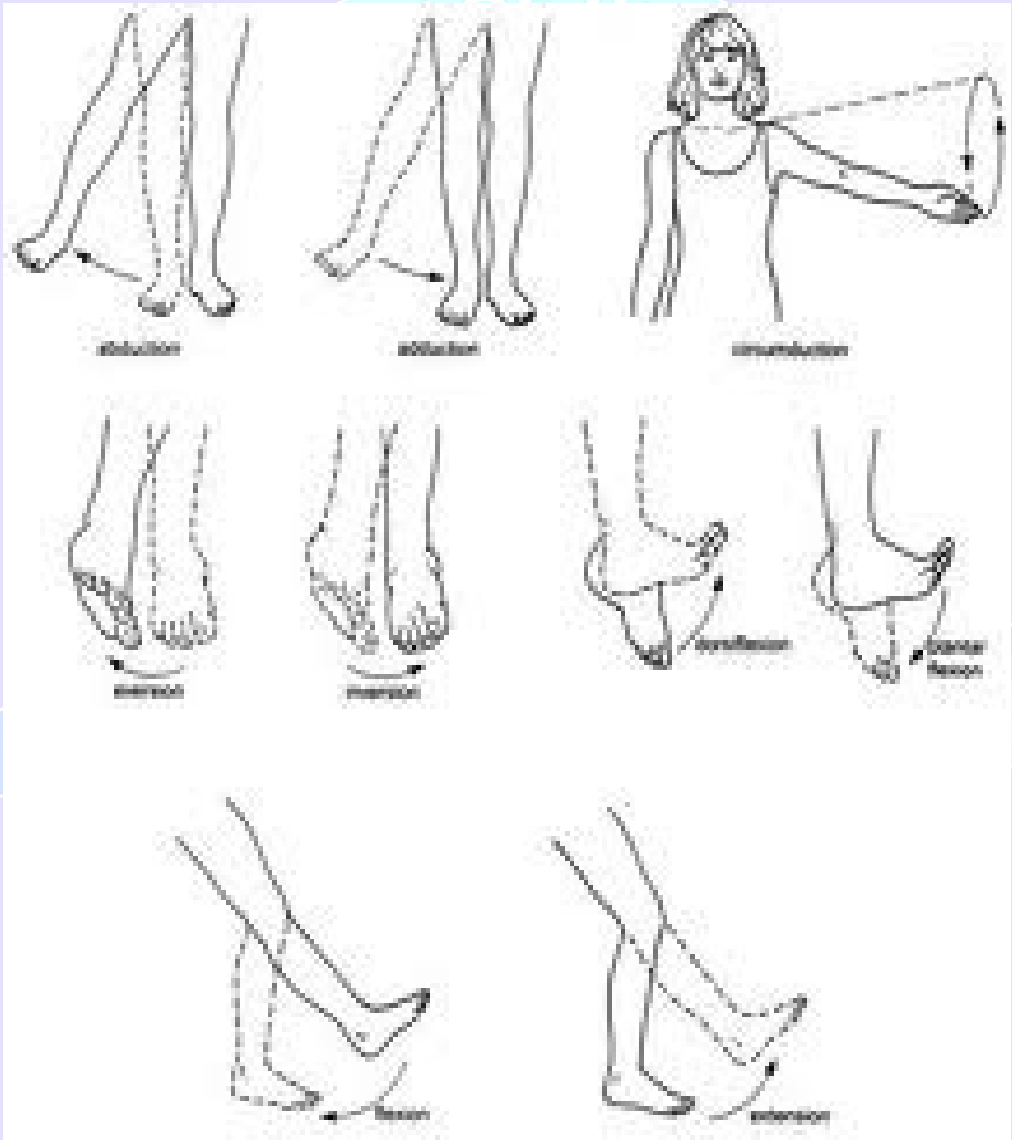


- The prefix **hyper-** is sometimes added to emphasize movement beyond the normal position, such as in **hyperflexion** or **hyperextension**.



- Such movements can put significant stress on the joints

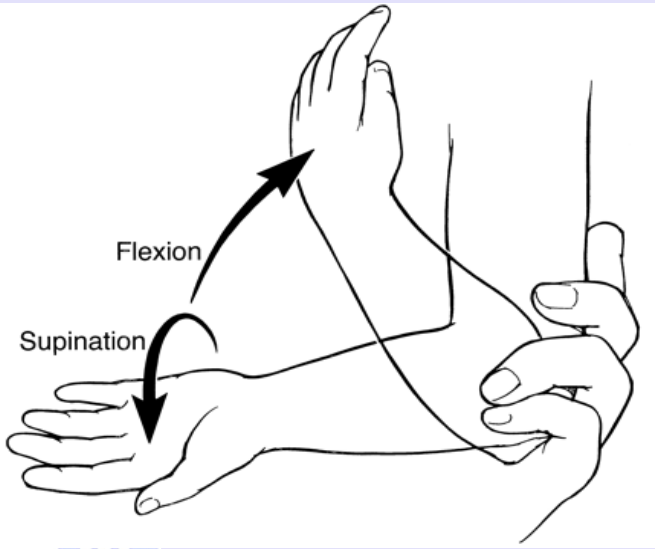
GENERAL MOTION

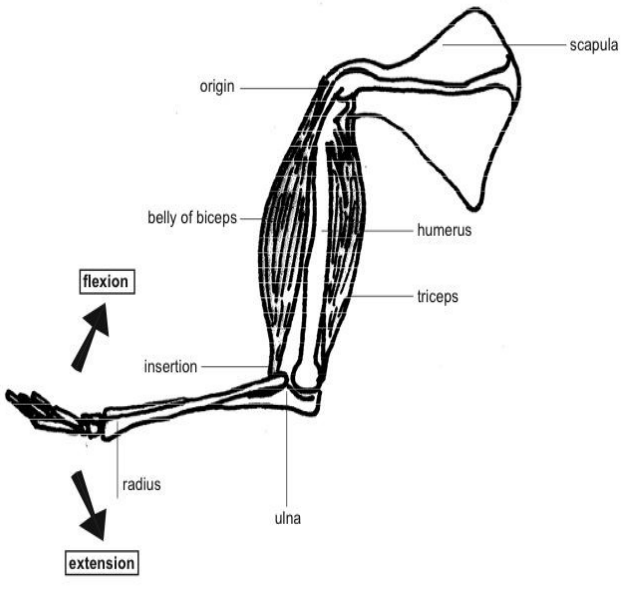


ADJUSTING ANGLE BETWEEN TWO PARTS

FLEXION

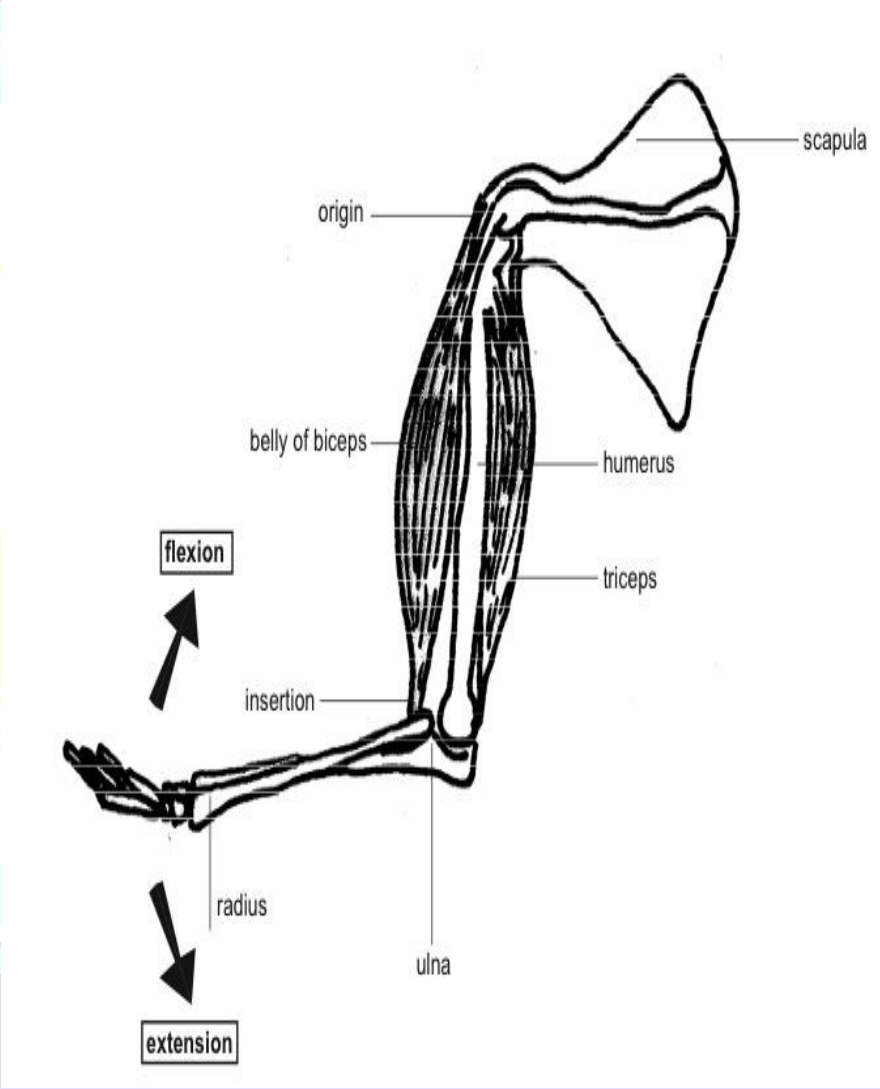
- Bending movement that decreases the angle between two parts.
- Bending the elbow, or clenching a hand into a fist, are examples of flexion.
- When sitting down, the knees are flexed.
- Flexion of the hip or shoulder moves the limb forward (towards the





EXTENSION

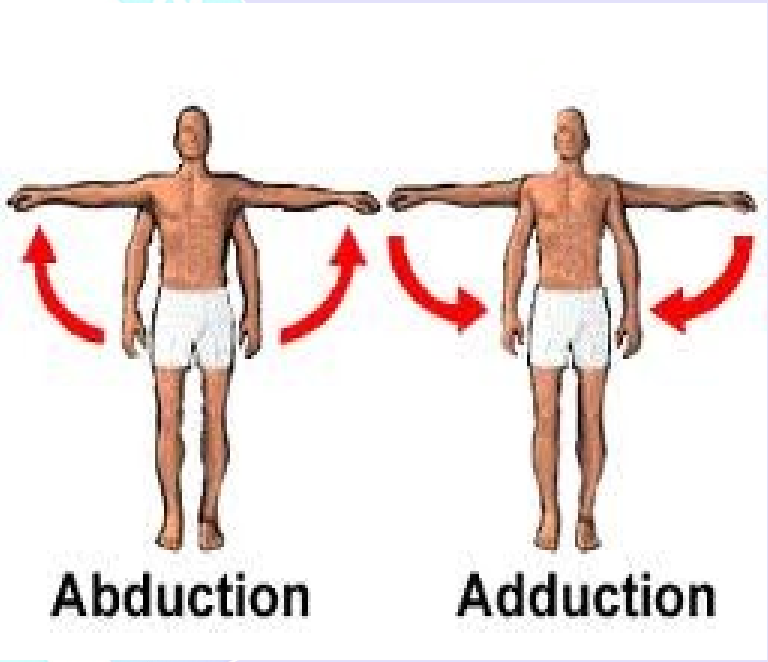
- The opposite of flexion; a straightening movement that *increases* the angle between body parts.
- In a conventional handshake, the fingers are fully extended.
- When standing up, the knees are extended.
- Extension of the hip or shoulder moves the limb backward (towards the posterior side of the body).



ADJUSTING RELATION TO MID-LINE OF BODY

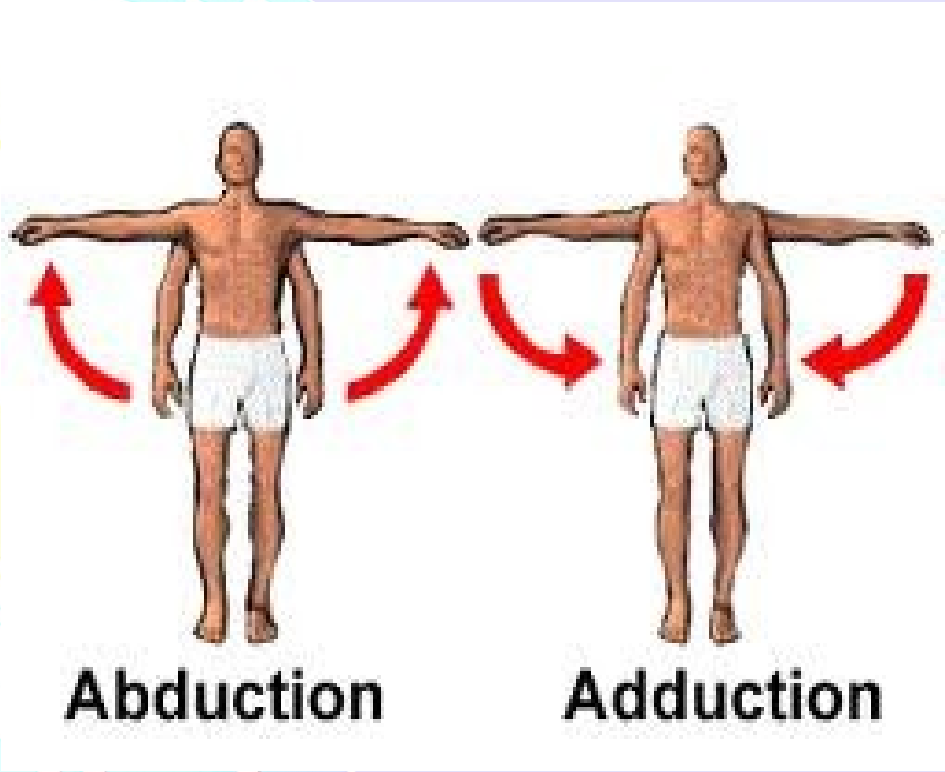
ABDUCTION

- A motion that pulls a structure or part away from the midline of the body (or, in the case of fingers and toes, spreading the digits apart, away from the centerline of the hand or foot).
- Abduction of the wrist is called radial deviation.
- Raising the arms laterally, to the sides, is an example of



ADDUCTION

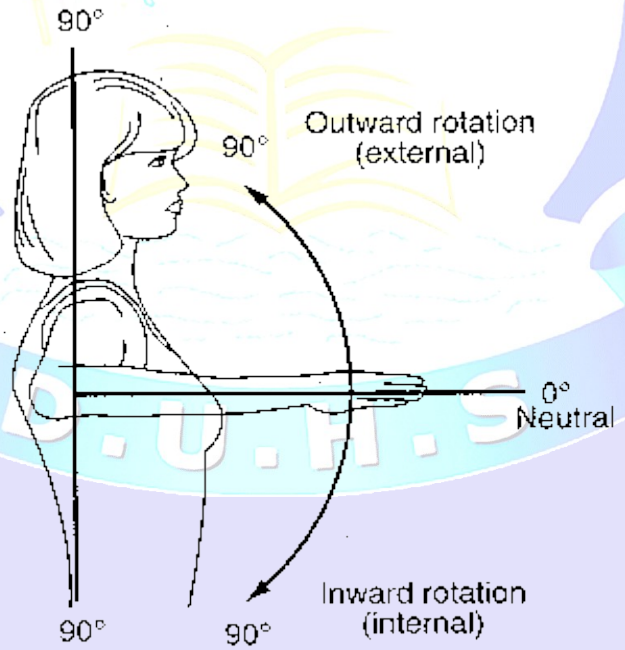
- A motion that pulls a structure or part towards the midline of the body, or towards the midline of a limb.
- Dropping the arms to the sides, or bringing the knees together, are examples of adduction.
- In the case of the fingers or toes, adduction is closing the digits together.
- Adduction of the wrist is called ulnar ***deviation.***



ROTATING BODY PARTS

INTERNAL ROTATION

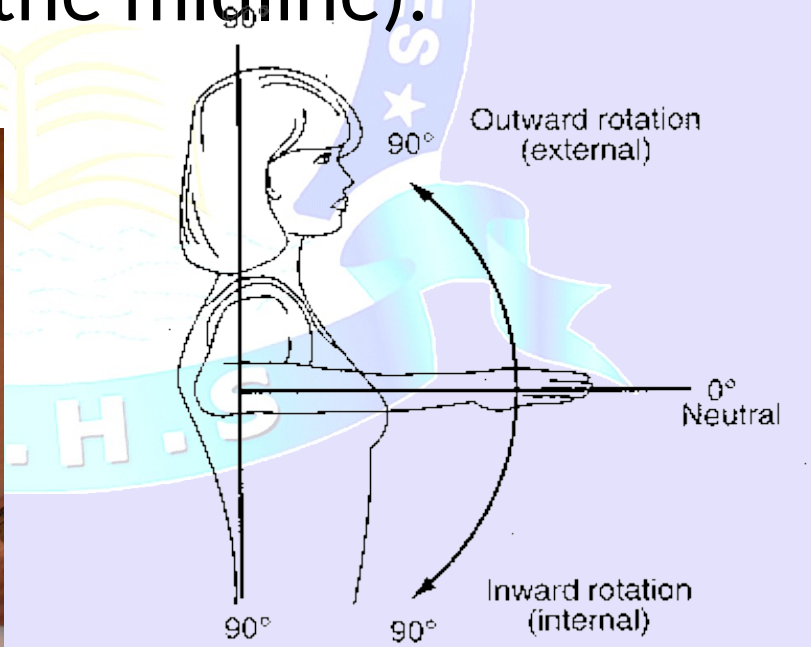
- Internal rotation (or medial rotation) of the shoulder or hip would point the toes or the flexed forearm inwards (towards the midline).



EXTERNAL ROTATION

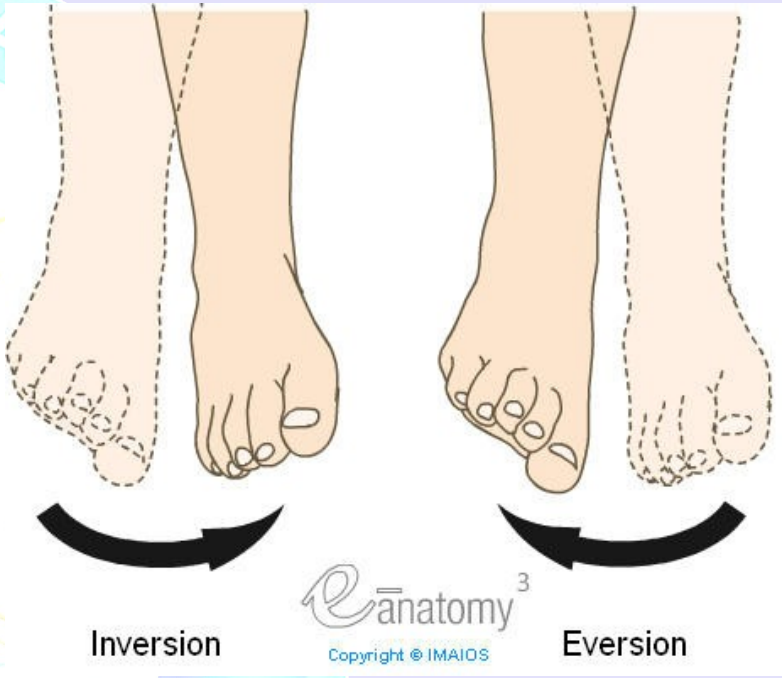
- External rotation (or lateral or outward rotation) is the opposite of Internal Rotation.
- It would turn the toes or the flexed forearm outwards (away from the midline).





EVERSION

- the movement of the sole of the foot away from the median plane



INVERSION

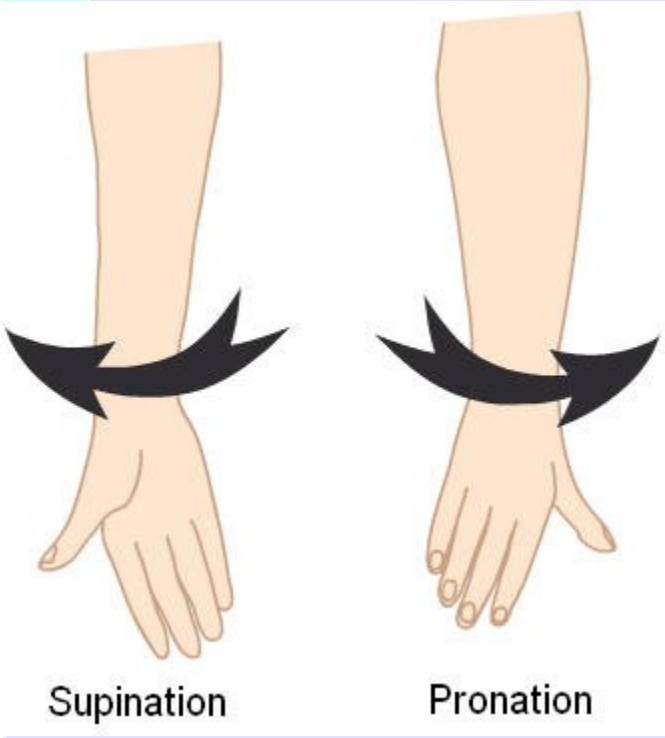
- The movement of the sole towards the median plane (same as when an ankle is twisted).

PRONATION

- **Pronation** is a rotational movement of the forearm at the radioulnar joint.

SUPINATION

- **Supination** is a position of forearm ; in the forearm when the palm faces anteriorly, or faces up (when the arms are unbent and at the sides).



OTHER SPECIAL MOTIONS

ANTERIOR/POSTERIOR MOVEMENT - GENERAL

- ***Protrusion*** ***- The anterior movement of an object. This term is often applied to the jaw.***

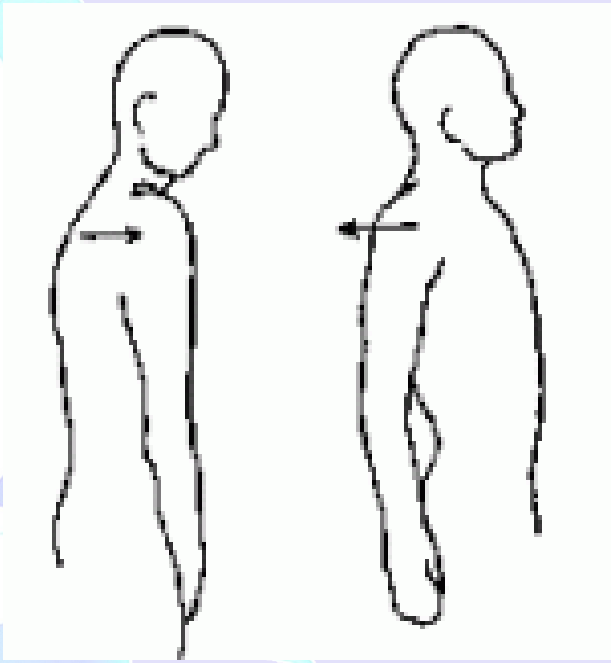
- ***Retrusion*** ***- The opposite of protrusion, moving a part posteriorly.***



ANTERIOR/POSTERIOR MOVEMENT - SHOULDERS

PROTRACTION:

- Anterior movement of the arms at the shoulders.



RETRACTION:

- Posterior movement of the arms at the shoulders.

CIRCUMDUCTION

- The circular (or, more precisely, conical) movement of a body part, such as a ball-and-socket joint or the eye.
- It consists of a combination of flexion, extension, adduction, and abduction.
- "Windmilling" the arms or rotating the hand from the wrist are



APPOSITION

- A motion involving a grasping of the thumb and fingers.



REPOSITION

- To release an object by spreading the fingers and thumb.

REFERENCES

- KLM Human Anatomy.




Thank You!